

## Language model

1. Consider the training set:

The Arabian knights

These are the fairy tales of the east

The stories of the Arabian knights are translated in many languages

Compute using the bigram model the probability of the sentence. Include start and end symbol in your calculations.

The Arabian knights are the fairy tales of the east

Soln:

The test sentence is  
The Arabian knights are the fairy tales of the east

$$\begin{aligned}P(\text{The}|\langle s \rangle) &= \frac{2}{3} \\P(\text{Arabian}|\text{The}) &= \frac{C(\text{The, Arabian})}{C(\text{The})} = \frac{1}{2} = 0.5 \\P(\text{knights}|\text{Arabian}) &= \frac{2}{2} = 1 \\P(\text{are}|\text{knights}) &= \frac{1}{2} \\P(\text{the}|\text{are}) &= \frac{1}{2} \\P(\text{fairy}|\text{the}) &= \frac{1}{2} = 0.33 \\P(\text{tales}|\text{fairy}) &= \frac{1}{1} \\P(\text{of}|\text{tales}) &= \frac{1}{1} = 1 \\P(\text{the}|\text{of}) &= \frac{2}{3} \\P(\text{east}|\text{the}) &= \frac{1}{3}\end{aligned}$$

So ans is obtained by multiplying all above

$$\begin{aligned}&= \frac{2}{3} \times \frac{1}{2} \times \frac{2}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times 1 \times 1 \times \frac{2}{3} \times \frac{1}{3} \\&= \frac{1}{162} = 0.0061728395.\end{aligned}$$

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2. You are an English class teacher and to make your course interesting you hold a write like Shakespeare competition where you ask each student to write a play in the style of Shakespeare. You try to score the plays automatically by using a trigram model where the probability distribution for the trigrams is

calculated using all of Shakespeare's plays as the corpus. While scoring the student plays however, you find that the data was not enough and most sentences in the student's plays have a score of 0. What options do you have to come out of your predicament if you still want to score the plays automatically?

Ans:

The options are the same as when we have inadequate language models. We back-off to simpler models e.g. bigram and unigram models and use smoothing to compute probabilities where the corpus does not have instances of the relevant n-gram. Another option is to use a weighted linear combination of multiple n-gram models for different values of n (e.g.  $n=1, 2, 3$ ).

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## Part of speech tagging and HMM

1. Using Penn Tree bank, find the POS tag sequence for the following sentences: [6 Marks]

1. The actor was happy he got a part in a movie even though the part was small. [2 marks]
2. I am full of ambition and hope and charm of life. But I can renounce everything at the time of need [3 marks]
3. When the going gets tough, the tough get going. [ 1 mark]

| Tag   | Description           | Example                | Tag  | Description           | Example              |
|-------|-----------------------|------------------------|------|-----------------------|----------------------|
| CC    | coordin. conjunction  | <i>and, but, or</i>    | SYM  | symbol                | <i>+, %, &amp;</i>   |
| CD    | cardinal number       | <i>one, two, three</i> | TO   | “to”                  | <i>to</i>            |
| DT    | determiner            | <i>a, the</i>          | UH   | interjection          | <i>ah, oops</i>      |
| EX    | existential ‘there’   | <i>there</i>           | VB   | verb, base form       | <i>eat</i>           |
| FW    | foreign word          | <i>mea culpa</i>       | VBD  | verb, past tense      | <i>ate</i>           |
| IN    | preposition/sub-conj  | <i>of, in, by</i>      | VBG  | verb, gerund          | <i>eating</i>        |
| JJ    | adjective             | <i>yellow</i>          | VCN  | verb, past participle | <i>eaten</i>         |
| JJR   | adj., comparative     | <i>bigger</i>          | VBP  | verb, non-3sg pres    | <i>eat</i>           |
| JJS   | adj., superlative     | <i>wildest</i>         | VBZ  | verb, 3sg pres        | <i>eats</i>          |
| LS    | list item marker      | <i>1, 2, One</i>       | WDT  | wh-determiner         | <i>which, that</i>   |
| MD    | modal                 | <i>can, should</i>     | WP   | wh-pronoun            | <i>what, who</i>     |
| NN    | noun, sing. or mass   | <i>llama</i>           | WP\$ | possessive wh-        | <i>whose</i>         |
| NNS   | noun, plural          | <i>llamas</i>          | WRB  | wh-adverb             | <i>how, where</i>    |
| NNP   | proper noun, singular | <i>IBM</i>             | \$   | dollar sign           | <i>\$</i>            |
| NNPS  | proper noun, plural   | <i>Carolinas</i>       | #    | pound sign            | <i>#</i>             |
| PDT   | predeterminer         | <i>all, both</i>       | “    | left quote            | <i>‘ or “</i>        |
| POS   | possessive ending     | <i>’s</i>              | ”    | right quote           | <i>’ or ”</i>        |
| PRP   | personal pronoun      | <i>I, you, he</i>      | (    | left parenthesis      | <i>[, (, {, &lt;</i> |
| PRP\$ | possessive pronoun    | <i>your, one’s</i>     | )    | right parenthesis     | <i>], ), }, &gt;</i> |
| RB    | adverb                | <i>quickly, never</i>  | ,    | comma                 | <i>,</i>             |
| RBR   | adverb, comparative   | <i>faster</i>          | .    | sentence-final punc   | <i>. ! ?</i>         |
| RBS   | adverb, superlative   | <i>fastest</i>         | :    | mid-sentence punc     | <i>: ; ... --</i>    |
| RP    | particle              | <i>up, off</i>         |      |                       |                      |

### Soln:

The/DT actor/NN was/VB happy/JJ he/PRP got/VB a/DT part/NN in/IN a/DT movie/NN “even though”/CC the/DT part/NN was/VB small/ADV. [2 marks]

I//PRP am/VB full/JJ of/IN ambition/NN and/CC hope/NN and/CC charm/JJ of/IN life/NN. But/CC I/PRP can/VB renounce/VB everything/JJ at/IN the/DT time/NN of/IN need/NN [3 marks]

When/WDT the/DT going/NN gets/VB tough/RB, the/DT tough/NN get/VB going/RB. [ 1 mark]

2.

**Part of speech tagging using Hidden Markov Model (HMM) – M2**

The following three sentences (S1, S2 and S3) and their corresponding tag sequences (T1, T2 and T3) are given as training data for implementing HMM. **Answer questions A-D.**

|                          |                                 |                                  |
|--------------------------|---------------------------------|----------------------------------|
| S1: John cut the paper . | S2: Mary asked for a hair cut . | S3: Sharon asked for a pay cut . |
| T1: N V D N STOP         | T2: N V I P N N STOP            | T3: N V I P N N STOP             |

What will be size of Emission and Tag translation matrices?

**Note: Include the start and the stop symbols and assume we are working with a bigram model.**

What will be the Emission Probability  $P(\text{asked} | V)$  ?

What will be the Tag Translation Probability  $P(I | V)$  ?

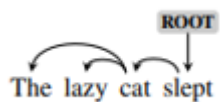
If a brute force approach is employed to find the tags for the test sentence "I had a deep cut.", how many possible tag sequences need to be evaluated?

## Parsing

1. Build a parse tree for the sentence "She loves to visit Goa" using Probabilistic Parsing [5marks]

S → NP VP 1.0  
VP → V PP 0.4  
VP → V NP 0.6  
PP → P NP 1.0  
NP → NP PP 0.3  
NP → N 0.3  
N → visit 0.3  
V → visit 0.6  
N → Goa 0.3  
N → She 0.5  
V → loves 1  
P → to 1  
DT → a 1

- 2) Give the correct sequence of arc eager parsing operations for the given sentence [2marks]



Solution:

|                   |                      |         |
|-------------------|----------------------|---------|
| [Root, ]          | [The lazy cat slept] | []      |
| [Root,The]        | [ lazy cat slept]    | [Shift] |
| [Root ,the ,Lazy] | [cat slept]          | [Shift] |
| [Root ,the ,Lazy] | [cat slept]          | [LA]    |
| [Root the ]       | [cat slept]          | [ LA]   |
| [Root,Cat]        | [slept]              | [SH]    |
| [Root, ]          | [ slept]             | [LA]    |
| [Root,Slept]      | []                   | [RA]    |

SH,SH,LA,LA,SH,LA,RA

3. Given the grammar and lexicon below dervie the parse tree using top down parsing method for the sentence [3 marks]

S :The guy ate pizza

**S->NP VP**

**VP->VNP**

**NP->Det N**

**N->pizza**

**N->guy ,Det ->the**

**V->ate**

**Soln:**

1The 2 guy 3 ate 4 the 5pizza 6

| State          | Backup State | Action  |
|----------------|--------------|---------|
| 1.((S) 1)      |              |         |
| 2.((NP VP) 1)  |              |         |
| 3.(DT N VP) 1) |              | matches |
| the            |              |         |
| 4.((N VP) 2)   |              | matches |
| guy            |              |         |
| 5.((VP)3)      |              |         |

6.((V NP ) 3)

ate

matches

7.(( Det N) 4)

matches the

8.((N ))5

matches pizza

9.()

4. Given the grammar and lexicon below show the final chart for the following sentence after applying the bottom up chart parser.[5 marks]

**S:** Book the flight on airasia

S->VP

VP->V NP

NP->NP PP

NP->Det Noun

PP ->Prep Noun

Det ->the

Verb->Book

Prep->on

Noun->flight|airasia

**Soln:**

S->VP

VP->Verb NP

VP->V NP

NP->NP PP

NP->Det Noun

PP->Prep Noun

| Verb   |   | Det |   | Noun   |   | preposition |   | Noun    |   |
|--------|---|-----|---|--------|---|-------------|---|---------|---|
| 0 Book | 1 | the | 2 | flight | 3 | on          | 4 | airasia | 5 |

VP->V.NP

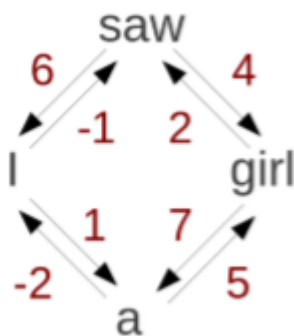
NP-> Det.Noun

NP->DetNoun.

PP->Prep Noun.

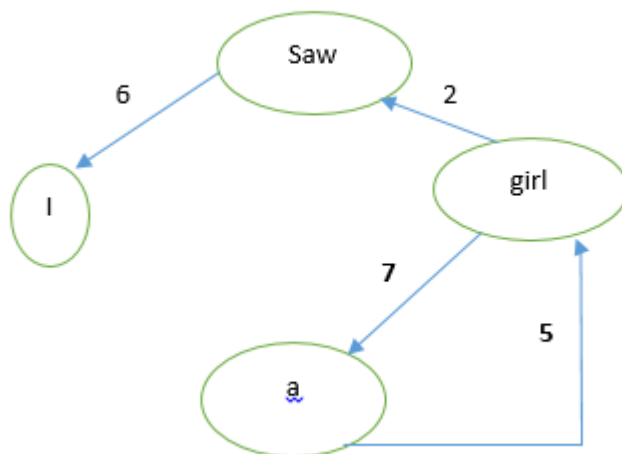
NP->NP. PP

6. In the below weighted graph, the edge weights between girl-saw and a-girl in the maximum spanning tree are: [5]



Soln:





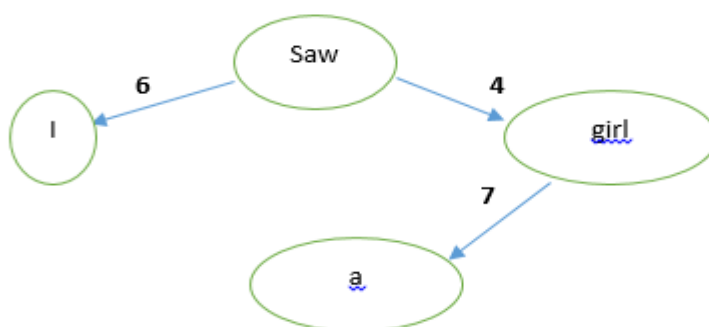
A  $\rightarrow$  girl  $\rightarrow$  forms a cycle

As per Chu Liu Edmond algorithm we can remove the cycle by contracting and expansion

Contracting step



Expanding



Because saw  $\rightarrow$  girl  $\rightarrow$  a gives more weightage than a  $\rightarrow$  girl  $\rightarrow$  saw

So the final weight for saw  $\rightarrow$  girl is 4 and a  $\rightarrow$  girl is 7

7. Given a treebank how would you determine probabilities for the grammar rules given in Question 1 (for use with a basic PCFG parser)?

Let's take the VP rule. There are three VP rules. I would count the total number of VP rules in the

Treebank. Then, for each rule, I would count the number of times that rule occurs and divide by the total number of VP rules. That would yield the probability for each rule. I would follow a similar procedure for each rule where the same non-terminal appeared on the left-hand side.

8.

1. Given the grammar and lexicon below (which is the same as that of question 5), show one possible **top-down derivation** for the sentence:

*Run the Detroit marathon*

|                                   |                                            |
|-----------------------------------|--------------------------------------------|
| $S \rightarrow NP VP$             | $Det \rightarrow the$                      |
| $S \rightarrow VP$                | $Noun \rightarrow run \text{ — } marathon$ |
| $NP \rightarrow Det NP$           | $Verb \rightarrow run$                     |
| $NP \rightarrow Proper-Noun Noun$ | $Proper-Noun \rightarrow Detroit$          |
| $VP \rightarrow Verb NP$          |                                            |

## Answer

$S \rightarrow VP$   
 $S \rightarrow Verb NP$   
 $S \rightarrow Verb Det NP$   
 $S \rightarrow run Det NP$   
 $S \rightarrow run the Proper-Noun Noun$   
 $S \rightarrow run the Detroit Noun$   
 $S \rightarrow run the Detroit marathon$

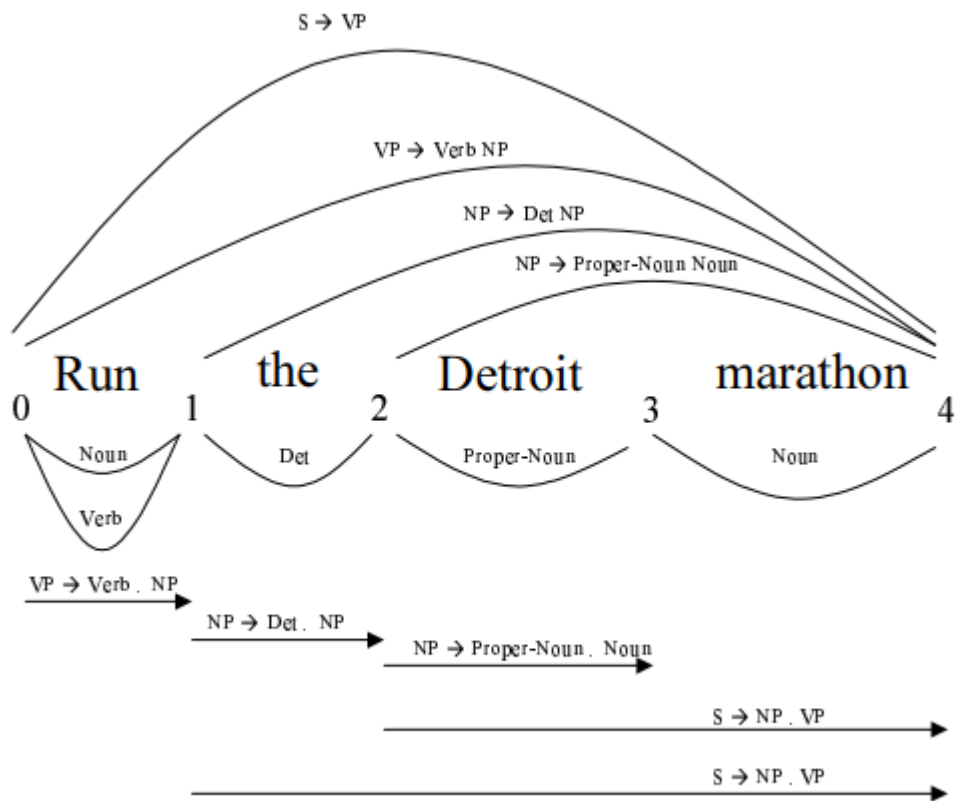
9.

Given the grammar and lexicon below, show the **final chart** for the following sentence after applying the bottom-up chart parser from class:

*Run the Detroit marathon*

Remember that the final chart contains all edges added during the parsing process. You may use either the notation from class (i.e. nodes/links) or the notation from the book to depict the chart.

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| $S \rightarrow NP VP$             | $Det \rightarrow the$                |
| $S \rightarrow VP$                | $Noun \rightarrow run \mid marathon$ |
| $NP \rightarrow Det NP$           | $Verb \rightarrow run$               |
| $NP \rightarrow Proper-Noun Noun$ | $Proper-Noun \rightarrow Detroit$    |
| $VP \rightarrow Verb NP$          |                                      |



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### Sentiment analysis

1. In this modern age where the internet is growing rapidly, the existence of the internet can make it easier for tourists to find information about hotels. Tourists usually tell the experience during the hotel by writing reviews on the internet. Hence many hotel's reviews are found on the internet. With the availability of reviews on the internet with large numbers, tourists can't understand all the reviews they read whether they contain positive or negative opinions. It takes a sentiment analysis to quickly detect if the reviews are positive or negative reviews. Using the Multinomial Naïve Bayes Classifier method find out that the given hotel reviews are positive or negative.

|    |                                                          |          |
|----|----------------------------------------------------------|----------|
| D1 | The hotel is clean and great                             | Positive |
| D2 | The hotel owner is very helpful                          | Positive |
| D3 | Overall Aston Hotel's experience was great               | Positive |
| D4 | The condition of the hotel was very bad                  | Negative |
| D5 | A HORRIBLE EXPERIENCE FOR ONE WEEK                       | Negative |
| D6 | The hotel view was great                                 | ?        |
| D7 | My holiday experience stay in usa so horrible            | ?        |
| D8 | Overall the hotel in aston very clean and great Positive | ?        |
|    |                                                          |          |

Soln:

| After smoothing | p(words   positive) | p(words   negative) |
|-----------------|---------------------|---------------------|
| word            | $\frac{1}{26}$      | $\frac{1}{22}$      |
| hotel           | $\frac{2}{26}$      | $\frac{1}{22}$      |
| clean           | $\frac{2}{26}$      | $\frac{1}{22}$      |
| great           | $\frac{3}{26}$      | $\frac{1}{22}$      |
| owner           | $\frac{2}{26}$      | $\frac{1}{22}$      |
| very            | $\frac{2}{26}$      | $\frac{1}{22}$      |
| helpful         | $\frac{2}{26}$      | $\frac{1}{22}$      |
| overall         | $\frac{2}{26}$      | $\frac{1}{22}$      |
| Asian           | $\frac{2}{26}$      | $\frac{1}{22}$      |
| experience      | $\frac{2}{26}$      | $\frac{1}{22}$      |
| conclusion      | $\frac{1}{26}$      | $\frac{1}{22}$      |
| Bad             | $\frac{1}{26}$      | $\frac{1}{22}$      |
| Horrible        | $\frac{1}{26}$      | $\frac{1}{22}$      |
| one             | $\frac{1}{26}$      | $\frac{1}{22}$      |
| week            | $\frac{1}{26}$      | $\frac{1}{22}$      |
|                 | $\frac{1}{26}$      | $\frac{1}{22}$      |

- 1)  $P(\text{Positive} | \text{sentence}) = 0.01$
- 2)  $P(\text{negative} | \text{sentence}) = 0.0016$   
 $D6 \rightarrow +ve$
- 3)  $P(\text{Positive} | \text{sentence}) = 0.0017$   
 $P(\text{negative} | \text{sentence}) = 0.0033$   
 $D1 \rightarrow -ve$
- 3)  $P(\text{Positive} | \text{sentence}) = 0.01$   
 $P(\text{negative} | \text{sentence}) = 0.0016$   
 $D8$  is positive

2. Given the following documents and their sentiment polarities

| Document | Sentiment words   | Polarity |
|----------|-------------------|----------|
| D1       | Good, Enjoy, Good | Positive |
| D2       | Poor, Unpleasant  | Negative |
| D3       | Enjoy ,Wonderful  | Positive |
| D4       | Good, Lovely      | Positive |
| D5       | Good, Poor, Rude  | Negative |
| D6       | Good ,Wonderful   | ?        |

Determine the sentiment polarity of document D6 using the multinomial naïve Bayes classification (with add1 smoothing) approach. Show your step in detail.

**Solution:**

**P (Positive) =3/5**

**P (Negative) =2/5**

P (Good /Positive) = $\frac{3+1}{7+7}=\frac{4}{14}$   
 $=\frac{1+1}{5+7}=\frac{2}{12}$

P (Good/Negative)

P (Wonderful/Positive) = $\frac{1+1}{7+7}=\frac{2}{14}$   
 (Wonderful/negative) = $\frac{0+1}{5+7}=\frac{1}{12}$

P

For the document 6

P (Positive/Good, Wonderful) =  $\frac{4}{14} \times \frac{2}{14} \times \frac{3}{5}$   
 $=0.29 \times 0.14 \times 0.6$   
 $=0.024$

P (Negative/ Good, Wonderful) = $\frac{2}{12} \times \frac{1}{12} \times \frac{2}{5}$   
 $=0.16 \times 0.083 \times 0.4$   
 $=0.005$

**Sentiment polarity of document D6 is Positive**

3. For sentiment analysis of twitter data, Which classifier did you choose among SVM and Naïve bayes and why? Justify your answer.[3m]

**Ans:**

**Option -1 mark**

**Justification (2marks)** since, you are given only the data of tweets and no other information, which means there is no target variable present. One cannot train a supervised learning model, both svm and naive bayes are supervised learning techniques.

**b) *I bought an iPhone a few days ago. It was such a nice phone. The touch screen was really cool. The voice quality was clear too. Although the battery life was not long, that is ok for me. However, my mother was mad with me as I did not tell her before I bought it. She also thought the phone was too expensive, and wanted me to return it to the shop.*** What are the problems associated with this type of sentiment analysis?[3Marks]

Ans:

Identification of Implicit aspects.

Multiple sentiments for same opinion phrase for different aspects

Association of corresponding sentiments to aspects in a multi-aspect review

4. How is sentiment calculated or scored? [2M]

5. Compare Rule based approach and machine learning approach in sentiment analysis [5].

6.

Given a list of positive and negative seed sentiment words and Table-1 providing the details of word occurrence and co-occurrences:

positive seeds (Pwords): good, nice, excellent, positive, fortunate, correct, superior

negative seeds (Nwords): bad, nasty, poor, negative, unfortunate, wrong, inferior

Compute the polarity of the phrase "excellent and outstanding".

| Word 1    | Word 2      | Count of word 1 | Count of word 2 | count of co-occurrences |
|-----------|-------------|-----------------|-----------------|-------------------------|
| excellent | outstanding | 5578            | 2749            | 1384                    |
| poor      | outstanding | 283891          | 3293296         | 3347                    |

**Table-1**

## **Machine translation**

1. Compute the BLEU score for the below translations (candidate1, candidate2). Consider 1gram, 2 gram, 3 gram, 4 gram and Brevity-Penalty for calculating BLUE score .

**Reference:** The teacher arrived late because of the traffic

**Candidate 1:** The teacher was late due to the traffic

**Candidate 2:** A teacher arrived late because of transportation

Soln:



Bleu Score.

candidate 1

Unigram  $= \frac{4}{7}$

Bigram  $= \frac{1}{6}$

Trigram  $= 0$

Four gram  $= 0$

candidate 2

Unigram  $= \frac{5}{7}$

Bigram  $= \frac{4}{6}$

Trigram  $= \frac{2}{5}$

four gram  $= \frac{1}{2}$

BP for candidate 1  $= 0.867$

Bleu score for candidate 1  $= 0$

BP for candidate 2  $= 0.615$

Bleu score for candidate 2  $= 0.571$

2. Compute the BLEU score for the below translations (candidate1, candidate2). Consider 1gram, 2 gram, 3 gram, 4 gram and **Brevity-Penalty for calculating BLUE score .**

**Reference:** The NASA Opportunity rover is battling a massive dust storm on Mars.

**Candidate 1:** The Opportunity rover is combating a big sandstorm on Mars.

**Candidate 2:** A NASA rover is fighting a massive storm on Mars.

Soln:

| Metric             | Candidate 1 | Candidate 2 |
|--------------------|-------------|-------------|
| precision1 (1gram) | 8/11        | 9/11        |
| precision2 (2gram) | 4/10        | 5/10        |
| precision3 (3gram) | 2/9         | 2/9         |
| precision4 (4gram) | 0/8         | 1/8         |
| Brevity-Penalty    | 0.83        | 0.83        |
| BLEU-Score         | 0.0         | 0.27        |

**3.** Compute the BLEU score for the machine translation output .

Assume 1gram ,2 gram and *brevity penalty*

**German:** Ich bin zur Zeit nicht im Büro  
**Reference English:** I am currently out of the office  
**MT English:** I am currently not in the office

Unigram

$$p_1 = \frac{1+1+1+0+0+1+1}{7} = \frac{5}{7} = 0.71$$

**Bigram**

English: (I am), (am currently), (currently out), (out, of), (of the), (the office)  
 MT English: (I am), (am currently), (currently not), (not in), (in the), (the office)

$$P1 = \frac{1+1+1+1+1+1}{6} = \frac{6}{6} = 1$$

*brevity penalty* = 1

$$\text{Final BLEU Score} = 1 * 0.71 * 0.67 = 0.48$$

4. Compute the BLEU score for the MT output.

**(Assume 1gram,2 gram.3 gram.4 gram and brevity penalty)**

Reference 1: The cat is on the mat.

Reference 2: There is a cat on the mat.

MT output: The cat the cat on the mat.

## Question and answer system and Chat bot

1. Identify the type of the dialogue agent from the following conversation. Also explain the working of that type of dialogue agent.

(5 marks)

System: How may I help you?

User: I want to book dinner at a restaurant that serves good Chinese food

System: Berkos is a restaurant that serves good Chinese food.

User: ok

System: Should I book a dinner?

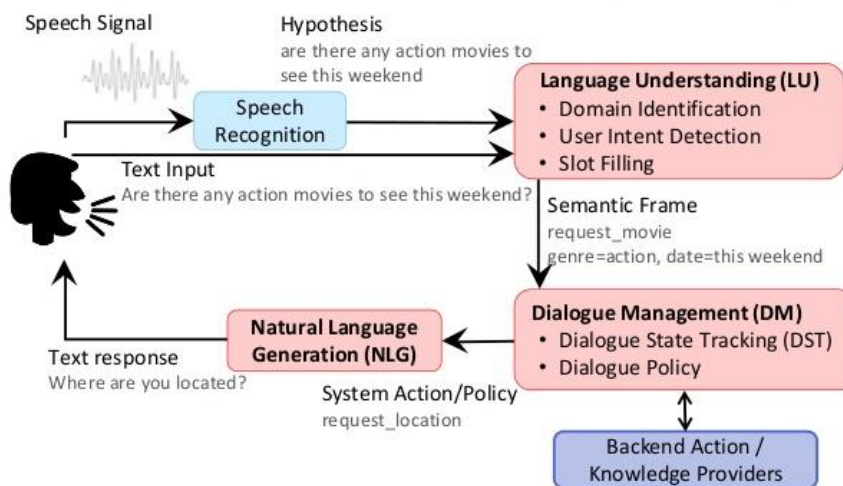
User: yes, please

System: ok.

**Soln:** Above is the task oriented dialogue system

The diagram shows the working of the task oriented dialogue system.

The student should explain briefly each step as mentioned in the slides



2. Find the intent, domain and slots for the following:

(2 marks)

Book an appointment on 12<sup>th</sup> Feb 2021 at 10 am for a ECG Test.

**SOLN:**

DOMAIN: Medical

INTENT: Book an Appointment

Slots

- Services: ECG TEST
- Date: 12<sup>th</sup> Feb 2021
- Time: 10 AM

3. In a collection of 10000 document, the following words occur in the following number of documents:

(3 marks)

Oasis occurs in 400 documents, Place occurs in 3500 documents, Desert occurs in 800 documents, Water occurs in 800 documents, Comes occur in 800 documents

Beneath occurs in 200 documents, Ground occurs in 900 documents

Calculate TF-IDF term vector for the following document:

Oasis Place Desert Water Comes Beneath Ground Place

| <del>Doc</del>                                                                   | (TF)<br>Term freq. | IDF                | TF * IDF |
|----------------------------------------------------------------------------------|--------------------|--------------------|----------|
| Oasis                                                                            | 1/8                | $\log(10000/400)$  | 0.1747   |
| Place                                                                            | 2/8                | $\log(10000/3500)$ | 0.11398  |
| Desert                                                                           | 1/8                | $\log(10000/800)$  | 0.137114 |
| Water                                                                            | 1/8                | $\log(10000/800)$  | 0.137114 |
| Comes                                                                            | 1/8                | $\log(10000/800)$  | 0.137114 |
| Beneath                                                                          | 1/8                | $\log(10000/200)$  | 0.212371 |
| Ground                                                                           | 1/8                | $\log(10000/900)$  | 0.13072  |
| TF-IDF vector (0.1747, 0.11398, 0.137114, 0.137114, 0.137114, 0.212371, 0.13072) |                    |                    |          |

## **Word sense disambiguation and ontology**

1) What are lexical sample task and all word task in word sense disambiguation? How can sources like Wikipedia be used for word sense disambiguation [2 marks]

### **Solution**

**What are lexical sample task and all word task in word sense disambiguation?**

**Lexical sample task and all word task are 2 variants of word sense disambiguation**

**? Lexical sample task -Small pre-selected set of target words**

**? All-words task - System is given an all-words entire texts and lexicon with an inventory of senses for each entry. We have to disambiguate every word in the text (or sometimes just every content word).**

2. How can sources like Wikipedia be used for word sense disambiguation  
Wikipedia can be used as training data for word sense disambiguation using supervised learning techniques

### **Ans:**

- **Concept is mentioned in a Wikipedia: article text may contain an explicit link to the concept's Wikipedia page, which is named by a unique identifier (can be used as a sense annotation)**

- **These sentences can then be added to the training data for a supervised system.**

3. How can WordNet relations be used for word sense disambiguation in following sentences:

[3 marks]

1. A bat is not a bird, but a mammal.

2. Jaguar reveals its quickest car ever

3. Raghuram Rajan was the 23rd Governor of the Reserve Bank of India

### **Solution**

Nouns and verbs can be extracted from the sentences. The senses in word net can be extracted for these words and senses with close relations can be extracted as correct sense.

1. Bat can be sports bat or mammal. But looking at nouns bat, bird and mammal, correct sense of bat as MAMMAL can be found using WordNet relations.

2. Jaguar can be a car or animal. Looking at nouns Jaguar, correct sense of Jaguar as CAR can be found using WordNet relations.

3. Bank can be river bank or financial bank. : Search senses of nouns Bank,"Raghuram Rajan",

Governor. The correct sense of BANK as FINANCIAL sense can be found using WordNet relations.

4. Consider below three types of movie reviews and convert it into Bag of words model:

- Review 1: This movie is very scary and long
- Review 2: This movie is not scary and is slow
- Review 3: This movie is spooky and good

|          | 1<br>This | 2<br>movie | 3<br>is | 4<br>very | 5<br>scary | 6<br>and | 7<br>long | 8<br>not | 9<br>slow | 10<br>spooky | 11<br>good |
|----------|-----------|------------|---------|-----------|------------|----------|-----------|----------|-----------|--------------|------------|
| Review 1 | 1         | 1          | 1       | 1         | 1          | 1        | 1         | 0        | 0         | 0            | 0          |
| Review 2 | 1         | 1          | 2       | 0         | 0          | 1        | 1         | 0        | 1         | 0            | 0          |
| Review 3 | 1         | 1          | 1       | 0         | 0          | 0        | 1         | 0        | 0         | 1            | 1          |

2.

- *the cat sat*
- *the cat sat in the hat*
- *the cat with the hat*

| Document                      | the | cat | sat | in | hat | with |
|-------------------------------|-----|-----|-----|----|-----|------|
| <i>the cat sat</i>            | 1   | 1   | 1   | 0  | 0   | 0    |
| <i>the cat sat in the hat</i> | 2   | 1   | 1   | 1  | 1   | 0    |
| <i>the cat with the hat</i>   | 2   | 1   | 0   | 0  | 1   | 1    |

## Semantic web

1. How is Syntactic web different from the Semantic web? What is URI in semantic web ontology?

Syntactic web consist of huge data on net connected by hyperlinks which is rendered by machines but machines cannot process it due to inability to understand the meaning of the content.

The semantic Web identifies a set of technologies, tools, and standards which form the basic building blocks of an infrastructure to support the vision of the Web associated with meaning.

A Universal Resource Identifier (URI) is a formatted string that serves as a means of identifying abstract or physical resource. A URI can be further classified as a locator, a name, or both. Every resource is identified with unique URI in ontology. Develop an OWL ontology using the following for animal kingdom for classes likecarnivorous, herbivorous and omnivorous. Use following Property characteristics, restrictions and Class expressions [3 marks]

❑ inverseOf

❑ domain

❑ range

❑ Cardinality

❑ disjointWith

❑ subClassOf



```

<rdf:Class rdf:ID="Carnivorous">
  <rdf:subClassOf rdf:resource="#Animal"/>
</rdf:Class>
<rdf:Class rdf:ID="Herbivorous">|
  <rdf:subClassOf rdf:resource="#Animal"/>
</rdf:Class>
<rdf:Class rdf:ID="Omnivorous">
  <rdf:subClassOf rdf:resource="#Animal"/>
</rdf:Class>

<rdf:Class rdf:ID="Carnivorous">
  <owl:disjointWith rdf:resource="#Herbivorous"/>
</rdf:Class>
<owl:Restriction>
  <owl:onProperty rdf:resource="#hasLegs" />
  <owl:cardinality
    rdf:datatype="&xsd:nonNegativeInteger">4</owl:cardinality>
</owl:Restriction>
<owl:ObjectProperty rdf:ID="Eats">
  <rdf:domain rdf:resource="#Carnivorous"/>
  <rdf:range rdf:resource="#Animal"/>
</owl:ObjectProperty>

```

1. Design a sample ontology for the 'academic university' domain. Clearly mention the
  - a. Classes
  - b. Properties
  - c. Relations
  - d. Axioms / constraints
 e.g. Professor, course with 'teaches' relation.

The ontology should contain about 10 classes with associated properties, relations and axioms and presented in RDF triple format.

#### Answer key:

Some samples below. Pls make sure that the student has mentioned at least 10 classes with associated properties, relations and axioms

Classes:

Professor      rdf:type owl:Class

Course          rdf:type owl:Class

Student rdf:type owl:Class

Research        rdf:type owl:Class

Publication      rdf:type owl:Class

Thesis            rdf:type owl:Class

Properties:

Name            rdf:type rdf:Property

Name            rdfs:domain      Professor

Name            rdfs:domain      Student

CourseID        rdf:type rdf:Property

CourseID        rdfs:domain      Course

Relations:

Professor        teaches Course

Publication      hasAuthor        Professor

Thesis            hasAuthor        Student

Professor        hasInterestArea      Research

Axioms:

Student owl:disjointWith      Professor

UndergradStudent      rdfs:subClassOf Student

GradStudent            rdfs:subClassOf Student

AssistantProfessor      rdfs:subClassOf Professor

AssociateProfessor      rdfs:subClassOf Professor

2. Describe any classification model (say, Naïve Bayes) to do word sense disambiguation (WSD). Clearly indicate what is the model, the features, how is it trained, and how it is used for prediction. Assume that disambiguation is for occurrence instances of a single word – for example, the occurrence of 'bank has different senses in the sentences below:

*The **bank** will not be accepting cash on Saturdays.*

*The river overflowed the **bank**.*

**Answer Key:**

For WSD, the context in which an ambiguous word occurs is represented by a vector of features  $F=(f_1, f_2, \dots, f_n)$  and the sense of the ambiguous word is represented by the class label  $(s_1, \dots, s_k)$ . Choosing the right sense of the ambiguous word  $w$  can be modeled as finding the sense  $s_i$  that maximizes the conditional probability  $P(w=s_i/F)$ .

$$\begin{aligned}
s' &= \arg \max_{s_i} P(w = s_i | F) \\
&= \arg \max_{s_i} \frac{P(F | w = s_i)}{P(F)} P(w = s_i) \\
&= \arg \max_{s_i} P(F | w = s_i) P(w = s_i)
\end{aligned}$$

The NB classifier assumes conditional independence of features, so that:

$$\begin{aligned}
s' &= \arg \max_{s_i} \prod_{f_j \in C} P(f_j | w = s_i) P(w = s_i) \\
&= \arg \max_{s_i} \left[ \sum_{f_j \in C} \log(P(f_j | w = s_i)) + \log P(w = s_i) \right]
\end{aligned}$$

The features for WSD using a NB algorithm are terms such as words, collocations, and words assigned with their positions which are extracted from the context of the ambiguous word. The probability of sense  $s_i$ ,  $P(s_i)$ , and the conditional probability of feature  $f_j$  with observation of sense  $s_i$ ,  $P(f_j | s_i)$ , are computed via Maximum-Likelihood Estimation:

$$\begin{aligned}
P(s_i) &= C(s_i) / N \\
P(f_j | w = s_i) &= C(f_j, s_i) / C(s_i)
\end{aligned}$$

Where  $C(f_j, s_i)$  is the number of occurrences of  $f_j$  in a context of sense  $s_j$  in the training corpus,  $C(s_i)$  is the number of occurrences of  $s_i$  in the training corpus, and  $N$  is the total number of occurrences of the ambiguous word  $w$  or the size of the training dataset. To avoid the effects of zero counts when estimating the conditional probabilities of the model, when meeting a new feature  $f_j$  in a context of the test dataset, for each sense  $s_i$  we set  $P(f_j | w = s_i)$  equal  $1/N$ .

3. If you are asked to build a statistical phrase-based machine translation model for a word order-based language (like, English), to a morphologically rich language (like, Hindi or Sanskrit),

- what problems do you foresee?
- how will you address them?

Answer Key:

One of the key problems arising during translation from English to a morphologically rich language (like, Hindi) is that of data sparsity.

There could be different strategies to address this –

1. collect large amounts of parallel corpora assuming that the data captures the morphological variations.
2. Other option could be to break the problem into ‘translation’ and then ‘morphology selection’. The source sentence / phrase is first translated into the target language with the root form of words. As a follow-up step appropriate inflection of the word is selected by a separate model.

4. You are designing a frame-based dialog system for movie booking.

a. What are the different slots in your design? Mention along with their corresponding entity types and questions that the system would ask a user.

b. Show a finite-state dialog manager for the system

c. What changes would you make to the design to change it from a single initiative system to multi-initiative system?

**Answer Key:**

| Slot       | Type  | Question                                    |
|------------|-------|---------------------------------------------|
| MOVIE NAME | Movie | Which movie do you want to watch?           |
| SHOW DATE  | Date  | What date you want to book the show for?    |
| SHOW TIME  | Time  | What time do you want to book the show for? |

Finite-state dialog manager – the system asks sequence of questions to fill up each slot. Each question is independent of the other.

Accept an answer that presents a sequence of questions. Refer to slide-8

[http://courses.washington.edu/ling575/SPR2017/slides/ling575\\_class4\\_DM\\_flat.pdf](http://courses.washington.edu/ling575/SPR2017/slides/ling575_class4_DM_flat.pdf)

The above system is single initiative as the control is always with the system. In multi-initiative systems, the control could arbitrarily shift between system/user. For this, we use the concept of frames that are collection of slots and their values. We then use models for entity extraction and slot filling on user responses. The dialog manager keeps track of which slots are filled up and which are available and then uses rule-based techniques to decide the next action (question).

5. “These earphones are a good pick at this price. Connected with laptop for office calls and these are working well although there is no noise cancellation. Quality of wires are a bit thin and look delicate, though neckband is ok. Bass will seem ok if you have not used good quality earphones earlier.”

You have been given product review data like the one shown above. You are asked to design a sentiment analysis model for this data. What would be your approach?

Describe the different components of your solution. State any assumptions that you are making and pros/cons (if any) of your approach.

**Answer Key:**

This product review data contains review of products and its different features.

Hence, an aspect-based sentiment analysis model would be most useful.

The model would have two key components –

1. An aspect extraction model to extract the product features. If the list of product attributes is already available, we could just build a dictionary-based extractor that does direct lookup in this dictionary. This would, however, be limited by the entries in the attributes dictionary and does not support semantic variations. In order to account for semantic variations, one could leverage word embedding- based models or use knowledge bases like WordNet. Although these models allow for semantic variations, they are still limited by the vocabulary of the attributes seen during training of these models. In order to support unseen attributes, one could train NER models.

2. A sentiment classification model for a given phrase / sentence. We could train a text classification model using n-gram features. This would require labeled data.

The model could be applied either at the sentence or phrasal level.